



SHARIF
UNIVERSITY OF
TECHNOLOGY

Foundations of Data Science Introduction (Part 1)

Sharif University of Technology
Department of Electrical Engineering

Instructor: Babak Khalaj
Fall 2026

Course Objectives

- Presenting basic ecosystem/workflow of applied data science and exposure to advanced concepts
- Review of fundamental math/statistics concepts required for practical data analysis
- Providing hands-on experience through application development for real-world data sets
- Working on large datasets and becoming familiar with development challenges
- Enhancing vision for understanding new technologies and applications in this field

Course Syllabus

1. Introduction to Data Science Ecosystem: A systems view
2. Basics of Data Models: Data Sources, Data Wrangling, Relational and Non-relations Databases, Streaming Data, Time Series Data
3. Introduction to database design and using entity relationship diagrams
4. Data Visualization
5. Data Preprocessing
6. Review of Statistical Analysis: Distributions, Hypothesis Testing, Sampling
7. Regression Through the Lens of Statistical Learning

Course Syllabus

7. Review of Supervised Machine Learning
8. Review of Unsupervised Machine Learning
9. Principles of Causality
10. Model Evaluation
11. Introduction to Deep Learning: Review of fundamental architectures e.g.,
CNN, RNN, LSTM, AE
12. Transformers: Foundations & Architectures

Course Syllabus

- 13. From Foundations to Frontiers: The Evolution of Generative Models
- 14. Modern Time-Series Modeling Using Transformers
- 15. Deployment: Methods for Deployment/Monitoring and their use case

Grading Policy

1. Homework (6 assignments): 7.5 points
2. Course Project: 4.5 points + 1 **bonus** point
3. Final Exam: 5 points
4. Teaching Assistant Quiz (4 quizzes): 3 points
5. Class Quiz: Floating **bonus** points! Be present to get a great grade!

Total: More than 20+1 points!

Pre Requisites

- Python programming and exposure to paradigms
- Probability and Statistics
- A course in Machine Learning either from our university or audited.
 - Knowledge of classical machine learning and exposure to deep learning is recommended.

We take a test to determine if you have the requirements for the course. This test is for your own understanding since we have assumed that everyone is the course is familiar with the concepts.

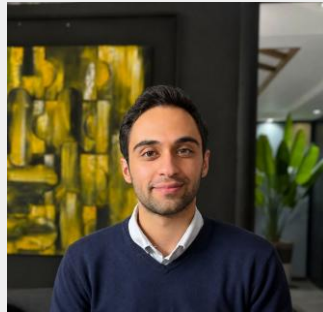
Teaching Assistant Team



**Mohammad
Eshtehardian**



Pooya Gholami



Kiarash Rashidi



**Mohammad
Hossein Momeni**



Amir Reza Azari



Shahab Hosseini



Sahar Semsarha



**Mohammad Ali
Naderi**



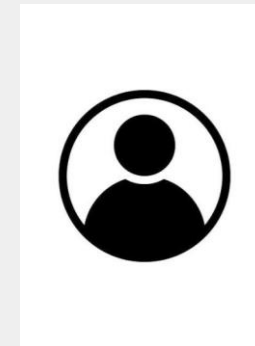
Darya Azaddel



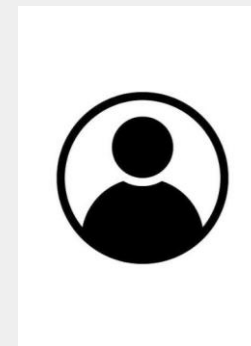
Ali Rezaei



Saba Atashfaraz



**Iman Alizadeh
Fakouri**



Aida Aryafar



**Hossein
Soleymani**

Teaching Assistant Sessions

The course will have regular T.A. sessions to cover the practical side of data science. The focus is more on employing **different tools to build standalone applications**. In addition, workshops for different tools will be held during the semester. Some are listed below:

1. Review of Python
2. Data Manipulation Libraries: NumPy, Pandas and Matplotlib
3. Exposure to databases: SQL
4. Applications in Production: Docker
5. Data Crawling : Beautiful Soup and Selenium
6. Scikit-learn
7. Visualization Libraries: Networkx and Plotly
8. Deep Learning Libraries: PyTorch and TensorFlow and etc.
9. Final Project Workshop

References

1. *Designing Machine Learning Systems*, Chip Huyen, O'Reilly Media, 2022
2. *Designing Data Intensive Applications*, Martin Kleppmann, O'Reilly Media, 2017
3. *Artificial Intelligence, A modern Approach*, Stuart J. Russell and Peter Norvig, Pearson, 2021
4. *Database Design for Mere Mortals*, Michael J. Hernandez, Addison-Wesley, 2003
5. *Computational and Inferential Thinking, The Foundations of Data Science*, Ani Adhikari and John DeNero, UC Berkeley, 2021.

References

6. *Introduction to Computation and Programming using Python: with application to computational modeling and understanding data*, John V. Guttag, The MIT Press, 2021
7. *An Introduction to Deep Generative Modeling*, Lars Ruthotto and Eldad Habor, 2021
8. *Deep Generative Models*, CS236 Stanford University, Fall 2021

The references are listed to show places you can find extra material for reading. They will be provided throughout the course.